

TASVIRLARNI TAHLIL QILISH VA ANGLASH · Amaliy mashg'ulot

4-amaliy mashg'ulot. Chegaralarni aniqlash

2026–2027 o'quv yili

Muhitni sozlash. Python 3.11 yoki 3.12, faqat CPU. Versiyalar qat'iy belgilangan (requirements.txt): numpy==2.4.4, scipy==1.17.1, pandas==3.0.2, scikit-learn==1.8.0, scikit-image==0.26.0, matplotlib==3.10.9, pillow==12.2.0, opencv-python-headless==4.13.0.92, torch==2.14.0, torchvision==0.29.0.

```
python -m venv .venv && source .venv/bin/activate # Windows: .venv\Scripts\activate
python -m pip install --upgrade pip && pip install -r requirements.txt
```

Mavzu. Diskret gradient (Prewitt, Sobel, Scharr), shovqin va silliqlash, Laplacian va Laplacian of Gaussian (LoG), Canny detektori (NMS, hysteresis), subpiksel joylashuv, Hough transform, chegara detektorini baholash (Precision, Recall, F1-score).

Maqsadlar. Talaba gradient moduli va yo'nalishini (atan2) qo'lda hisoblaydi; yadrolarning masshtabi va shovqin kuchaytirishini chiqaradi; Canny ning har bir bosqichini kichik misolda bajaradi va NumPy'da yozadi; Canny + Hough asosidagi amaliy zanjirlarni (hujjat skaneri, tanga sanash, sudoku to'ri) quradi va natijani sonli baholaydi.

Baholash (100 ball). 1–16-topshiriqlar (qo'lda hisoblash, keltirib chiqarish, nazariy savollar): har biri 2 ball (32). 17–30-topshiriqlar (kod): 26, 27, 28 (*, amaliy holatlar) — har biri 8 ball (24), qolgan 11 tasi — 4 ball (44). Kod xatosiz ishlashi, seed=42 bilan takrorlanishi va ko'rsatilgan formatda chiqarishi shart.

Kelishuv. u — ustun (gorizontal, o'ngga), v — qator (vertikal, pastga). Yadrolar *korrelyatsiya* sifatida qo'llanadi (cv2.filter2D, cv2.Sobel kabi):

$$S_u = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}, P_u = \begin{pmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{pmatrix}, H_u = \begin{pmatrix} -3 & 0 & 3 \\ -10 & 0 & 10 \\ -3 & 0 & 3 \end{pmatrix}, S_v = S_u^T, \theta = \text{atan2}(I_v, I_u).$$

A qism. Qo'lda hisoblash, keltirib chiqarish, nazariya

- Sobel.** 3×3 oyna $A = \begin{pmatrix} 10 & 30 & 60 \\ 20 & 50 & 100 \\ 20 & 60 & 110 \end{pmatrix}$. Markaziy pikselda $I_u, I_v, M = \sqrt{I_u^2 + I_v^2}$ va θ ni (gradusda, 0,01 gacha) toping. Chegara chizig'i qaysi yo'nalishda ketadi?
- Prewitt va Scharr.** 1-topshiriqdagi A uchun Prewitt va Scharr bilan I_u, I_v, M, θ ni hisoblang. Uch yadroning burchaklarini solishtiring.
- atan2 va arctan.** a) $(I_u, I_v) = (-120, 90)$ uchun $M, \theta = \text{atan2}(I_v, I_u), \arctan(I_v/I_u)$ ni toping. Chegara oriyentatsiyasi ($\theta \bmod 180^\circ$) va chegara chizig'ining yo'nalishi qanday? b) $(0, -50)$ uchun θ ; nega arctan bu yerda yaramaydi?
- Chiziqli qiyyalik.** $I(u, v) = 2u - 3v + 7$. Prewitt, Sobel, Scharr javoblari $(I_u, I_v), M, \theta$ ni toping va haqiqiy hosila $(2, -3)$ bilan masshtab koeffitsiyentlarini $(6, 8, 32)$ tekshiring.
- Ajraladiganlik (keltirib chiqarish).** H_u ni ikki vektorning tashqi ko'paytmasi sifatida yozing. $k \times k$ ajraladigan yadro uchun piksel boshiga ko'paytirishlar sonini (k^2 va $2k$) chiqaring ($k = 7$ da son qiymat). Ko'ndalang silliqlovchi $[a \ b \ a]$ ning chastota javobi $b + 2a \cos \omega$ ekanini ko'rsating va $\omega = \pi$ da Prewitt, Sobel, Scharr uchun normallangan qiymatini toping.
- Shovqin.** Oq shovqin $\mathcal{N}(0, \sigma^2)$, $\sigma = 8$. $\|S_u\|_2, \|H_u\|_2$; tekis sohada Sobel va Scharr modulining kutilmasi $\mathbb{E}M = \|w\|_2 \sigma \sqrt{\pi/2}$; soxta chegara ehtimoli $p = 10^{-4}$ bo'lishi uchun ostona $t = \|w\|_2 \sigma \sqrt{2 \ln(1/p)}$ (ikkala yadro uchun).
- Laplacian.** a) Oyna $B = \begin{pmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 90 \end{pmatrix}$ markazida L_4 va L_8 javobi. b) 1D qiya chegara $f = (20, 20, 30, 60, 90, 100, 100)$ ga $[1 \ -2 \ 1]$ ni qo'llang (*valid* rejim) va nolni kesib o'tish joyini toping.
- LoG (keltirib chiqarish).** $G_\sigma = \frac{1}{2\pi\sigma^2} e^{-r^2/(2\sigma^2)}$ dan $\Delta G_\sigma = \frac{r^2 - 2\sigma^2}{\sigma^4} G_\sigma$ ni chiqaring. LoG qaysi radiusda nolga teng? $\iint \Delta G_\sigma du dv = 0$ ekanini ko'rsating va bu tekis sohadagi javob uchun nimani anglatishini tushuntiring.
- NMS.** Modullar $\begin{pmatrix} 30 & 70 & 95 \\ 40 & 80 & 60 \\ 90 & 50 & 35 \end{pmatrix}$, markaz 80. To'rt sektorli NMS ($0^\circ, 45^\circ, 90^\circ, 135^\circ$; v pastga) bo'yicha $\theta = 0^\circ, 45^\circ, 90^\circ, 135^\circ, -30^\circ, 100^\circ$ uchun markaz saqlanadimi? Har birida taqqoslangan qo'shnilarni yozing.
- Hysteresis (2D).** NMS xaritasi $N = \begin{pmatrix} 0 & 45 & 0 & 0 & 0 & 0 \\ 0 & 0 & 120 & 0 & 0 & 60 \\ 0 & 50 & 0 & 0 & 70 & 0 \\ 0 & 0 & 0 & 0 & 0 & 90 \end{pmatrix}$, $t_{\text{past}} = 40$, $t_{\text{yuqori}} = 100$. 8- va 4-bog'lanish uchun yakuniy chegara piksellerini toping.

11. L_1 va L_2 modul. (I_u, I_v) : (200, 200), (280, 0), (150, -230), (-90, 300); ostona $t = 300$. Qaysilari L_2 va qaysilari L_1 modul bilan ostonadan oʻtadi? M_1/M_2 ning oraligʻini keltirib chiqaring.
12. Baholash. $|E^*| = 320$, $|E| = 400$; E ning 288 pikseli E^* ning 1 piksel atrofida, E^* ning 272 pikseli E ning 1 piksel atrofida. Precision, Recall, F1-score (4 xona).
13. Hough transform. $\rho = u \cos \varphi + v \sin \varphi$. Nuqtalar (1, 3), (2, 2), (4, 0), (3, 3) uchun $\varphi \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$ da ρ ni hisoblang. Akkumulyator maksimumi qayerda va u qaysi chiziqli? 640×480 tasvir uchun ρ oraligʻi va $\Delta\rho = 1$, $\Delta\varphi = 1^\circ$ da akkumulyator katakchalari soni.
14. Kichik Canny zanjiri. 5×6 tasvirning har bir qatori (10, 10, 20, 80, 100, 100), silliqlashsiz, $t_{\text{past}} = 100$, $t_{\text{yuqori}} = 200$. Ichki ustunlar ($u = 1..4$) uchun Sobel I_u , NMS, hysteresis natijasi; parabola bilan subpiksel siljish $\delta = \frac{M_- - M_+}{2(M_- - 2M_0 + M_+)}$ va chegara joyi. Uni $55 = (10 + 100)/2$ darajani chiziqli interpolyatsiya bergan joy bilan solishtiring.
15. Nazariy savollar. a) cv2.Canny Gauss silliqlashni bajaradimi? b) Nega cv2.Sobel(img, cv2.CV_8U, 1, 0) xato natija beradi? c) Gradient va chegara yoʻnalishi qanday bogʻlangan? d) Median $m = 120$, $s = 0,33$ boʻlsa avtomatik Canny ostonalari. e) Nega $t_{\text{yuqori}} : t_{\text{past}} \approx 2:1-3:1$?
16. Oʻtkirlashtirish. $f = (20, 20, 20, 100, 100, 100)$, chetlarda $\Delta f = 0$ deb oling. $\alpha = 0,25$ uchun $f - \alpha \Delta f$ ni toping. 2D yadro $K = \delta - \alpha L_4$ oq shovqinni necha marta kuchaytiradi ($\|K\|_2$)?

B qism. Dasturlash (Python, OpenCV, NumPy)

Barcha topshiriqlar uchun umumiy kod (skimage.data tasvirlari paket ichida, internet talab qilinmaydi; tashqi fayl faqat 26-topshiriqda):

```
import math, urllib.request
from pathlib import Path
import numpy as np, cv2
from scipy import ndimage as ndi
from scipy.special import ndtr
from skimage import data
SEED = 42; K8 = np.ones((3, 3), np.uint8)
URL = "https://raw.githubusercontent.com/opencv/opencv/4.x/samples/data/"

def fetch(name): # ./data/ ga yuklab, keshlaydi
    p = Path("data") / name; p.parent.mkdir(exist_ok=True)
    if not p.exists(): urllib.request.urlretrieve(URL + name, p)
    return p

def disk_scene(noise, seed=SEED): # 200x200: fon 70, disk (100,100), r=60, 170
    v, u = np.mgrid[:200, :200]
    mask = (u - 100)**2 + (v - 100)**2 <= 60**2
    img = np.where(mask, 170.0, 70.0)
    rng = np.random.default_rng(seed)
    noisy = np.clip(img + rng.normal(0, noise, img.shape), 0, 255).astype(np.uint8)
    m8 = mask.astype(np.uint8)
    return noisy, (cv2.dilate(m8, K8) - cv2.erode(m8, K8)) > 0 # (tasvir, GT chegara)

def f1_score(e, gt, tol=1): # tol piksel tolerantlik bilan (P, R, F1), 3 xona
    e = e > 0; k = np.ones((2*tol + 1,)*2, np.uint8)
    p = (e & (cv2.dilate(gt.astype(np.uint8), k) > 0)).sum() / max(e.sum(), 1)
    r = (gt & (cv2.dilate(e.astype(np.uint8), k) > 0)).sum() / gt.sum()
    f = 0.0 if p + r == 0 else 2*p*r / (p + r)
    return round(float(p), 3), round(float(r), 3), round(float(f), 3)

def n_parts(e): # 8-bogʻlangan boʻlaklar soni
    return cv2.connectedComponents((e > 0).astype(np.uint8), connectivity=8)[0] - 1
```

Har bir topshiriq alohida funksiya (t17_...(), ...) boʻlib, natijani koʻrsatilgan formatdagi satr(lar)da chiqaradi; rasmlar ./out/ ga saqlanadi. Hujjatlar: https://docs.opencv.org/4.x/d5/d0f/tutorial_py_gradients.html, https://docs.opencv.org/4.x/da/d22/tutorial_py_canny.html, https://docs.opencv.org/4.x/d6/d10/tutorial_py_houghlines.html, https://docs.opencv.org/4.x/da/d53/tutorial_py_houghcircles.html, https://docs.opencv.org/4.x/da/d6e/tutorial_py_geometric_transformations.html, https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.gaussian_laplace.html, <https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.label.html>, <https://scikit-image.org/docs/stable/api/skimage.data.html>.

17. Sobel NumPy'da. `sobel_manual(a: np.ndarray) -> tuple[np.ndarray, np.ndarray]: np.pad(a, 1, mode='reflect') (= BORDER_REFLECT_101) va kesimlar bilan (I_u, I_v) , sikllarsiz. Tasvir: data.camera().astype(np.float64). cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3) bilan maksimal farq, M ning oʻrtachasi, maksimumi va $M > 100$ ulushi. Chiqish: T17 max|gx-cv2| = ... max|gy-cv2| =`

... va T17 mean M = ..., max M = ..., ulush(M>100) =

18. **Oriyentatsiya gistogrammasi.** `data.camera()` uchun θ mod 180° ning M bilan tortilgan 6 binli (30°) normalangan gistogrammasi (`np.histogram(..., weights=M)`), 3 xona; eng katta bin. Soʻng tasvirni `np.rot90` bilan burib takrorlang. Gistogramma qanday siljidi va nega?
19. **Yadrolarning burchak aniqligi.** 81×81 Gauss dogʻi $e^{-(u^2+v^2)/(2 \cdot 10^2)}$ (markaz (40, 40)), haqiqiy yoʻnalish $\text{atan2}(-v, -u)$. `cv2.filter2D` bilan Prewitt, Sobel, Scharr; $3 < r < 25$ halqada burchak xatosining oʻrtachasi va maksimumi (gradus, 4 xona). Nisbatlarni izohlang.
20. **Shovqindan soxta gradient.** `rng = np.random.default_rng(42)`; ketma-ket $\sigma = 5, 10, 20$ uchun `f = 128.0 + rng.normal(0, s, (400, 400))`. Sobel moduli oʻrtachasi (chetdan 2 piksel tashlab) va $\sigma\sqrt{6\pi}$; $t = \sigma\sqrt{24 \ln 1000}$ dan katta ulush; `GaussianBlur(f, (0,0), 1.5)` dan keyingi oʻrtacha (chetdan 8 piksel tashlab). Chiqish: T20 `sigma=...: nazariy=.. simul=.. t=.. ulush(M>t)=.. silliq(1.5)=..`
21. **Oʻz Canny.** `my_canny(img, lo, hi) -> np.ndarray[bool]: cv2.Sobel` (L_2 modul), 4 sektorli NMS (sektor $((\text{ang}+22.5)/45) \% 4$; bir qoʻshniga qatʼiy $>$, ikkinchisiga $\geq$), chetki piksellar 0, `ndi.label` (8-bogʻlanish) bilan hysteresis. Kirish: `cv2.GaussianBlur(data.camera(), (0,0), 1.4)`, (50, 150). `cv2.Canny(..., L2gradient=True)` bilan piksellar soni, aniq Jaccard va oʻz piksellaringizning 1 piksel ichida mos kelgan ulushi; rasm `out/t21_canny_compare.png`.
22. **Hysteresis taʼsiri.** 21-topshiriqdagi silliqlangan tasvirda `cv2.Canny` (L_2) ni (150, 150), (50, 50), (50, 150) ostonalari bilan ishga tushiring: piksellar, boʻlaklar (`n_parts`), boʻlakka piksel (2 xona).
23. **Avtomatik Canny.** `auto_canny(img, s=0.33) -> (lo, hi, edges): m = median, lo=int(max(0, (1-s)m)), hi=int(min(255, (1+s)m))`. Tasvirlar: `camera`, `coins`, `astronaut` (kulrang, `COLOR_RGB2GRAY`), avval `GaussianBlur(..., (0,0), 1.0)`. Median, ostonalar, chegara piksellari ulushi (4 xona).
24. **Marr–Hildreth.** `log_edges(img, sigma, thr) -> np.ndarray[bool]: ndi.gaussian_laplace`, oʻng yoki pastki qoʻshni bilan ishora oʻzgarishi va $|\text{farq}| > \text{thr}$ (chegara chap/yuqori pikselga yoziladi). `disk_scene(10)` da $\sigma = 2$, $\text{thr} \in \{0, 2, 4\}$; `f1_score` bilan. Solishtirish: Gauss(2) + Canny(50, 150).
25. **Shovqin boʻyicha F1.** `disk_scene(s)`, $s = 0, 10, 20, 30, 40$: xom Canny(50, 150) va `GaussianBlur((0,0), 2.0)` + Canny(50, 150) (ikkalasi L_2); F1 va boʻlaklar soni. Qaysi darajadan xom Canny ishlamay qoladi?
26. **★ Sudoku toʻri (Hough).** `fetch('sudoku.png')` (<https://raw.githubusercontent.com/opencv/opencv/4.x/samples/data/sudoku.png>), kulrang. `GaussianBlur((0,0), 1.5)`, Canny(50, 150) L_2 , `cv2.HoughLines(e, 1, np.pi/180, 120)`. Chiziqlar soni; $|\theta - 90^\circ| \leq 10^\circ$ (gorizontal), $\theta \leq 10^\circ$ yoki $\geq 170^\circ$ (vertikal); eng kuchli 3 ta (ρ, θ) ; chiziqlarni `out/t26_sudoku_hough.png` ga chizing. Nega chiziqlar soni toʻrdagi 20 ta chiziqqa teng boʻlsa ham ularning bir qismi takror?
27. **★ Hujjat skaneri (Canny + Hough).** Sunʼiy hujjat:

```
def make_document(seed=SEED):
    page = cv2.resize(data.page(), (768, 382))
    page = np.clip(page.astype(np.float64)*0.5 + 120, 0, 255).astype(np.uint8)
    h, w = page.shape
    src = np.float32([[0, 0], [w-1, 0], [w-1, h-1], [0, h-1]])
    dst = np.float32([[110, 95], [545, 70], [585, 380], [80, 410]]) # haqiqiy burchaklar
    canvas = cv2.warpPerspective(page, cv2.getPerspectiveTransform(src, dst), (640, 480), borderValue=40)
    rng = np.random.default_rng(seed)
    return np.clip(canvas + rng.normal(0, 6, canvas.shape), 0, 255).astype(np.uint8), dst
```

Zanjir: `GaussianBlur((0,0), 2.0) → Canny(30, 90) L2 → HoughLines(e, 1, np.pi/180, 120) → ovozlar tartibida takror boʻlmagan 4 chiziq ($|\Delta\theta| < 10^\circ$ va $|\Delta\rho| < 30$ boʻlsa takror; $\theta \approx 0$ va $\approx 180^\circ$ holatida ρ ishorasi almashadi) → kesishmalar ($30^\circ < |\Delta\theta| < 150^\circ$ juftliklar) → burchaklarni tartiblash (chap-yuqori, oʻng-yuqori, oʻng-past, chap-past) → warpPerspective (768 × 382). Chiqish: tanlangan (ρ, θ) , burchaklar (1 xona), har bir burchak xatosi va maksimumi (px); out/t27_document_flat.png.`

28. **★ Tangalarni sanash (HoughCircles).** `data.coins()`, `cv2.medianBlur(g, 3)`, `cv2.HoughCircles(b, cv2.HOUGH_GRADIENT, dp=1, minDist=25, param1=150, param2=22, minRadius=15, maxRadius=35)`. Tangalar soni, radius oʻrtachasi (2 xona), min, max; aylanalarni `out/t28_coins_hough.png` ga chizing. `param2` ni 20–30 oraligʻida oʻzgartirib, son qanday oʻzgarishini kuzating.
29. **Fokus oʻlchovi.** $F = \text{var}(cv2.Laplacian(a, cv2.CV_64F, ksize=1))$. `camera` (`float64`) uchun Gauss xiraligi $\sigma = 0, 1, 2, 3$ da F (2 xona). $\sigma = 2$ tasvirni $b - \alpha \Delta b$ ($\alpha = 0,5; 1,0, [0, 255]$ ga qirqib) bilan oʻtkirlashtirib, F ni qayta hisoblang.
30. **Subpiksel chegara.** `row = 50 + 100*ndtr((u - u0)/1.0), u = 0..39, np.tile(row, (10, 1))`; $u_0 \in \{20,0; 20,25; 20,5; 20,75\}$. 5-qatordagi Sobel I_u ning argmaksimumi k , parabola siljishi δ , baho $k + \delta$ va xato (4 xona). Chiqish: T30 `u0=...: argmax=.. delta=.. baho=.. xato=...`